

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

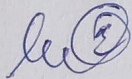
Annexure A

Descriptive Written Test

Code of Conduct:

I K.R. Jeevan Balaji (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



16
20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home automation System

Introduction:

An Intelligent Agent is a system that perceives its environment using sensors and takes action using actuators to achieve specific goals.

Types: 1) Simple Reflex Agent: Acts only on the current input using predefined condition action rules.

Working: Motion detected $\rightarrow$ Turn on lights, No motion $\rightarrow$ Turn off light.

Advantage: * Fast response, easy to design, low cost

Disadvantage: No memory, Cannot learn user habit, Fails in complex situation

2) Model Based Reflex Agent: Maintains an internal model of the environment and uses past information.

Working: * Remembers which rooms are occupied, Detects if a door was left open.

Advantages: * Uses current and past info, Better decision making works in partially observable environment.

Disadvantage: More complex, Requires memory.

3) Goal-Based Agent: choose action that help achieve a specific goal

Working: * Maintain comfortable temperature, keep house secure, Save electricity

Ex: if the goal is energy saving, the agent turn off unnecessary appliances even if nobody gives command.

Advantage: Feasible, Finds action that achieve goal, Better planning.

Disadvantage: Requires search algorithm, slower than reflex agent.

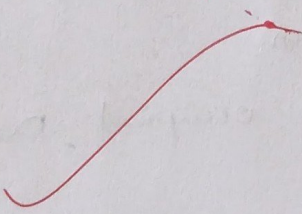
→ Utility Based Agent: selects the actions that provides the highest overall benefit

working: comfort, energy efficiency, security, electricity cost

Advantage: Makes optimal decision, handles conflicting goals, Best for real-life applications.

Disadvantage: Utility calculation is complex; Requires more computation

Conclusion: Hybrid intelligent agent is most effective choice because it remembers past information, plans according to goals and selects the actions with maximum overall benefits.



Robot is trying to escape from an unknown maze - complex network of paths

→ the robot does not know the maze layout it has to explore different paths until it find the exit

Uniform cost Search (UCS)

It is an uninformed search strategie

uniform cost search expands the lowest path cost

it uses priority queue.

Ex: Path A $\rightarrow$ cost = 10

Path B $\rightarrow$ cost = 50

ucs use path B because it has the lowest cost
it uses more memory

Iterative Deeping Search (IDS)

it is also an uninformed search strategie

IDS repeatedly perform DFS with increasing depth

it uses stack

Ex: Depth 0 $\rightarrow$ Depth 1 $\rightarrow$ Depth 2 $\rightarrow$ Depth 3

Does

It continues until find the exit

it uses less memory.

Time and Space Complexity

for UCS:

Time complexity: $O(b^{(c^*/d)})$

Space complexity: $O(b^{(c^*/d)})$

for IDS:

Time complexity: $O(b^{d^*})$

Space complexity: $O(b^{d^*})$

The best model would be Goal Based approach and Uniformed cost Search but if we want it to consume ~~less~~ memory IDS would be best.